Predicting Dementia Using Brain Scan Measurements and Patient Health Data

Healthcare Analytics Mini Project (CA-III) | Symbiosis Institute of Technology, Nagpur

THE PROBLEM

Dementia is usually spotted with a brain scan or a memory test, but the two are often looked at separately. This project asks: if brain scan measurements and patient details are put into one model, does it tell demented and nondemented people apart better, and which kind of information does it actually rely on?

THE DATA

- **OASIS-2** (training): 150 people aged 60 to 98, 373 visits.
- **OASIS-1** (second test set): 198 records of people aged 60 and above who have a CDR score.
- **Label:** demented or nondemented (OASIS-2 group label with 'converted' counted as demented; OASIS-1 CDR above 0).
- Classes are nearly balanced (51% / 49%), so no oversampling was needed. Missing SES (19) and MMSE (2) values were filled with the median.

TWO KINDS OF DATA, ONE MODEL

Brain scan measurements

eTIV: size of the skull cavity
nWBV: how much of it the brain fills
ASF: scaling factor, worked out from eTIV

Patient details

Age and **Sex**
SES: socioeconomic status
MMSE: score on a 30-point memory test

Education was left out because it is coded differently in OASIS-1 and OASIS-2.

HOW IT WAS DONE

1. Explore and clean the data, check missing values and class balance.
2. Split by person: all visits of one person stay together, so nobody is in both train and test.
3. Compare Logistic Regression, Random Forest and SVM (5-fold, repeated with 5 seeds).
4. Compare imaging only, clinical only and combined features.
5. Explain the Random Forest with SHAP.
6. Test on OASIS-1, a different group of people.
7. Deploy as a Streamlit dashboard.

WHAT IS NEW HERE

- SHAP influence is added up separately for brain scan measurements and for clinical details, to show how much of the prediction comes from each side. The closest paper I reviewed used SHAP on a mixed table without this split.
- Splitting by person matters. A plain random split inflated the scores (Random Forest AUC about 0.91, against about 0.79 when each person's visits stay together).
- Tested on a second dataset, then deployed as a live dashboard that shows a SHAP chart for each person.

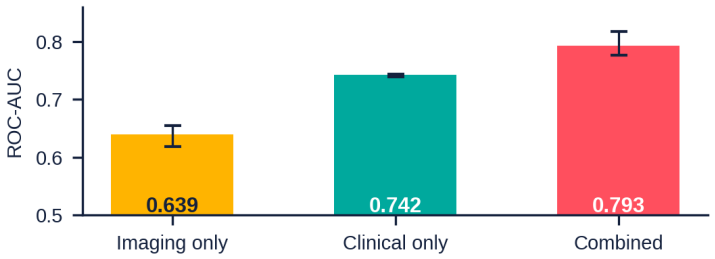
RESULTS

0.793
AUC with both kinds of data (imaging only 0.639, clinical only 0.742)

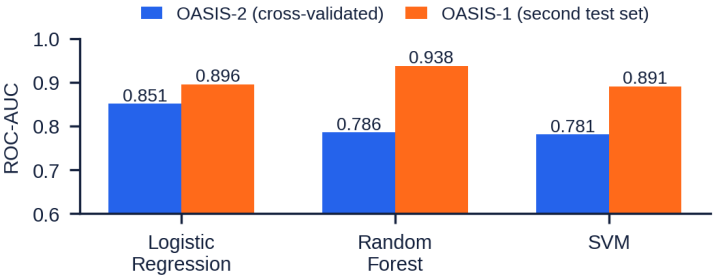
34% / 66%
share of SHAP influence: imaging / clinical

0.938
AUC on OASIS-1 with the Random Forest

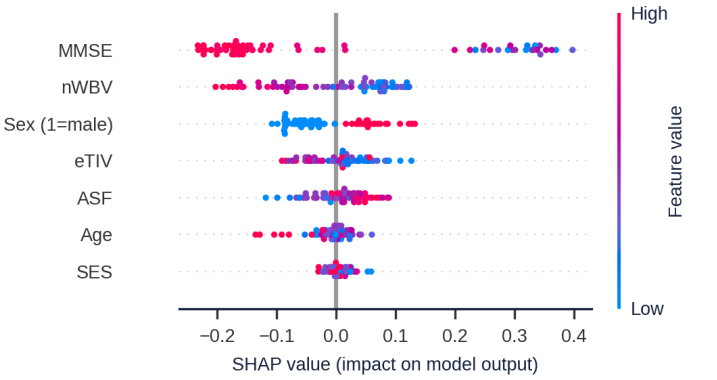
0.851
best AUC on OASIS-2 (Logistic Regression)



Both kinds of data together beat either alone (average of 5 seeds, whiskers = lowest and highest seed).



Logistic Regression leads on OASIS-2, Random Forest on OASIS-1. No single model wins everywhere.



SHAP summary, each dot is one visit. Low MMSE and low brain volume (nWBV) push towards demented. Imaging features carry 34% of the total influence on average (31% to 37% across 5 splits).

WHAT THE RESULTS SAY

- MMSE has the biggest influence in every split. It is closely linked to how the dementia label is assigned, so this is expected.
- Clinical details carry about two thirds of the influence, brain measurements about one third.
- On OASIS-2 the models miss about 3 in 10 demented cases (recall about 0.7).
- Small dataset (150 people), ages 60 and above only. SHAP shows what the model used, not what causes dementia.

CONCLUSION

Combining brain scan measurements with patient details predicted dementia status better than either one alone (AUC 0.79, against 0.64 and 0.74). The Random Forest also worked on a second dataset (AUC 0.94 on OASIS-1). Most of the prediction comes from the clinical details.

FUTURE SCOPE

More people and other datasets, a decision threshold tuned to catch more cases, extra measurements, and testing on follow-up visits.

LIVE DASHBOARD

Enter brain scan measurements and patient details to get a risk estimate and a chart of what pushed it. Built with Streamlit.

Open the app:
dementia-risk-predictor-e5ozaf75khdibrrdpxkfqk.streamlit.app

Code on GitHub:
github.com/chirag-2025/dementia-risk-predictor

A learning project, not a medical tool.

References. [1] Faouri, AlBashayreh and Azzeh (2022), Decision Science Letters. [2] Kubi and Nazir (2024), Int. J. of Information Technology. [3] Dhariwal (2026), Frontiers in Computational Neuroscience. [4] Basheer, Bhatia and Sakri (2021), IEEE Access.
Data. OASIS-1 and OASIS-2 (oasis-brains.org); the CSV copy used here came from Kaggle. **Tools.** Python, pandas, scikit-learn, SHAP, matplotlib, Streamlit, Google Colab, GitHub.

Chirag Somkuwar
PRN 23070521042